SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I A.Sravanthi / 192250871 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Sravanthi
Signature of the student:

95/100

Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message.
b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-by-step.
b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

a. Analyze how investigators could detect the presence of hidden data within the image.
b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25 20	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30 30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20 20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15 15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10 10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

95/100

A

CO1-AT1

Application Level

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Reg No :- 192325071

Course Name : Cryptography And Network Security

Course Code :- CSA5101

1) Scenario event communication security : A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

a) Reconstruct how the encrypted message might appear if a shift of 4 positions is applied

The caesar cipher is a substitution cipher in which each letter of the plaintext is replaced by another letter located a fixed number of positions ahead in the alphabet.

plaintext : MEET AT NOON

key : 4

This means every letter is moved 4 positions forward

Formula for caesar cipher encryption

$E(P) = (P+K) \cdot 1.26$

Where :

P = plaintext letter value

K = key

1.26 = Number of letters in the english alphabet

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14

P	Q	R	S	T	U	V	W	X	Y	Z
15	16	17	18	19	20	21	22	23	24	25

E(M) = (P+K) %.26

= (12+4) %.26 = 16 → Q

E(E) = (P+K) %.26

= (4+4) %.26 = 8 → I

E(E) = (P+K) %.26

= (4+4) %.26 = 8 → I

E(T) = (P+K) %.26

= (19+4) %.26 = 23 → X

E(A) = (P+K) %.26

= (0+4) %.26 = 4 → E

E(T) = (P+K) %.26

= (19+4) %.26 = 23 → X

E(N) = (P+K) %.26

= (13+4) %.26 = 17 → R

E(O) = (P+K) %.26

= (14+4) %.26 = 18 → S

$E(O) = (P+K) \% 26$

$= (14+4) \% 26 = 18 \rightarrow S$

$E(N) = (P+K) \% 26$

$= (13+4) \% 26 = 17 \rightarrow R$

Encrypted Message :- @11XEXRSSR

b) Explain how an authorized member, aware of the encoding rule, would retrieve the original message

Formula for decryption

$D(C) = (C-K) \% 26$

Where:

P = plaintext value

C = ciphertext value

K = key value

$D(@) = (C-K) \% 26$

$= (16-4) \% 26 \rightarrow 12 \rightarrow M$

$D(1) = (C-K) \% 26$

$= (8-4) \% 26 = 4 \rightarrow E$

$D(1) = (C-K) \% 26$

$= (8-4) \% 26 = 4 \rightarrow E$

$D(X) = (C-K) \% 26$

$(23-4) \% 26 = 19 \rightarrow T$

$D(E) = (C-K) \% 26$

$= (4-4) \% 26 = 0 \rightarrow A$

$D(X) = (C-K) \% 26$

$= (23-4) \% 26 = 19 \rightarrow T$

$D(E) = (C-K) \% 26$

$= (17-4) \% 26 = 13 \rightarrow N$

$D(S) = (C-K) \% 26$

$= (18-4) \% 26 = 14 \rightarrow O$

$D(S) = (C-K) \% 26$

$(18-4) \% 26 = 14 \rightarrow O$

$D(E) = (C-K) \% 26$

$= (17-4) \% 26 = 13 \rightarrow N$

Original message :- MEET AT NOON

plaintext : MEET AT NOON

Ciphertext : QIIX EX RSSR

2. Scenario : secure Messaging system A company deploys a keyword-based encryption technique suspect that a company for internal communication. An employee sends the word HECLO using the keyword "KEY" to ensure confidentiality

a) Demonstrate how the encryption process transform the message step by step

Cipher used :- The encryption technique used is the vigenere Cipher.

The vigenere cipher is a polyalphabetic substitution cipher that uses a keyword to encrypt plaintext. Each letter of the plaintext is shifted according to the corresponding letter of the keyword.

$E_i = (P_i + e_i) \% 26$

P_i = plaintext value

e_i = keyword value

mod26 = Number of letter in the english alphabet

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

Plaintext = HELLO

Key = KEY

plaintext	H	E	L	L	O	
plaintext value(p)	7	4	11	11	14	
key	K	E	Y	K	E	
key value (k)	10	4	24	10	4	
Ciphertext value (c)	17	8	35	21	18	
Ciphertext	R	I	J	V	S	

b) Explain how the repeating keyword influences each character of the ciphertext

The keyword "KEY" is repeated as "KEYKE" to match the length of the plaintext

*Each keyword letter provides a different shift value

keyword letter	shift value
K	10
E	4
Y	24
K	10
E	4

Because the shift changes for each position, the same plaintext letter can produce different ciphertext letters

Ex:- H is shifted by K (10 positions) and becomes R

E is shifted by E (4 positions) and becomes I

L is shifted by Y (24 positions) and becomes J